

Abhishek Sanjay Gadavi

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Career Objective

Aspiring VLSI Engineer with hands-on experience in CMOS circuit design, LTspice simulations, and Verilog-based digital design. Seeking an opportunity to contribute to Analog and Digital IC design while expanding my expertise in semiconductor technology.

Technical Skills

- **Programming:** C, Java, Python
- **Core:** Analog Electronics, CMOS VLSI Design, Semiconductor Devices, , Digital Electronics
- **Tools:** LTspice, MATLAB, Xilinx Vivado, Arduino IDE
- **Hardware:** Circuit Design
- **Soft Skills:** Problem Solving, Analytical Thinking, Team Collaboration

Projects

- **Two-Stage CMOS Operational Amplifier Design**
 - Designed and simulated a two-stage CMOS operational amplifier using 180 nm CMOS technology in LTspice for low-power analog applications.
 - Implemented a differential input stage with current mirror biasing, active load configuration, second gain stage, and Miller compensation to achieve high gain and stable frequency response.
 - Performed DC, AC, and transient simulations to evaluate key performance parameters including DC gain, gain-bandwidth product (GBW), phase margin, slew rate, input common-mode range, and output voltage swing.
 - Optimized transistor dimensions (W/L ratios) and compensation capacitor values to improve amplifier stability, frequency response, and power efficiency while minimizing propagation delay.
 - Verified circuit functionality by analyzing input-output characteristics, frequency response, transient behavior, and power dissipation .
 - Compared simulation results with design specifications and improved gain, bandwidth, and overall analog performance.
- **4-bit Ripple Carry Adder Design**
 - Designed and implemented a 4-bit Ripple Carry Adder using CMOS logic by cascading four 1-bit full adder circuits.
 - Developed transistor-level CMOS full adders using PMOS and NMOS transistors,
 - Simulated the complete design in LTspice using 180 nm CMOS technology to verify addition of multi-bit binary inputs under different test conditions.
 - Performed transient analysis to study carry propagation delay and evaluated its impact on the overall speed of the ripple carry architecture.
 - Analyzed power consumption, propagation delay, and output waveforms to assess circuit performance .
 - Optimized transistor sizing to achieve improved speed, lower power dissipation, and reliable logic operation while maintaining design stability.
- **IoT-Based Smart Energy Meter**
 - Designed and developed an IoT-based Smart Energy Meter for real-time monitoring of electrical energy consumption.
 - Integrated Arduino/ESP8266 with voltage and current sensors to measure electrical parameters and transmit data wirelessly.
 - Implemented cloud connectivity to enable remote monitoring and visualization of energy usage through a web or mobile application.
 - Performed real-time acquisition and analysis of voltage, current, power, and energy consumption to improve monitoring accuracy.
 - Developed a cost-effective solution for efficient energy management, reducing manual meter reading and enabling timely usage insights.

Education

University of Visvesvaraya College of Engineering	2023 – 2027
<i>B.Tech in Electronics and Communication Engineering (CGPA: 9.42)</i>	<i>Bengaluru</i>
Bapuji Independent PU College	2023
<i>PCMB – Karnataka State Board (97.80%)</i>	<i>Badagandi</i>
Bapuji International School	2021
<i>CBSE (94.60%)</i>	<i>Badagandi</i>

Trainings & Certifications

- **Infosys Training (Summer Internship I)** – Problem Solving, Communication and Leadership.
- **Samsung Innovation Campus (SIC)** – Coding & Programming Certification (2024).
- **Campus Champ (Summer Internship II)** – Data Structures and Algorithms Training.
- **Embedded Systems Training** – Successfully completed at BMS College of Engineering (BMSCE), Bengaluru.